

Requirements Traceability Matrix (RTM)

Legend:

- BR: Business Requirement
- FR: Functional Requirement
- DC: Design Component
- US: User Story / Feature
- TC: Test Case
- TS: Test Scenario
- CM: Code Module

Definitions:

- Requirement ID: Unique identifier for requirement
- Requirement Description: Description
- Business Priority: High, Medium, Low
- Source: BRD/Stakeholder
- Functional Area: Process area impacted
- Design Component: Architecture/module/solution mapping
- User Story/Feature Mapping: Agile/Story ticket ID
- Code Module/Repository Reference: Source code mapping
- Test Scenario ID: High-level test group
- Test Case ID: Atomic test
- Test Status: Not Started/In Progress/Pass/Fail
- Dependencies: Upstream requirements
- Risks: Associated project/product risks
- Comments: Additional notes

Requirement ID	Requirement Description	Business Priority	Source	Functional Area	Design Component	User Story/Feature Mapping	Code Module/Repository Reference	Test Scenario ID	Test Case ID	Test Status	Dependencies	Risks	Comments
BR-001	Read source csv data from volume and load to delta table customer and order	High	BRD	Data Ingestion	ETL Pipeline	US-001	repo/etl/load_csv.py	TS-001	TC-001	Not Started			Infrastructure setup Data source inaccessible

| BR-002 | Add 'TotalAmount' column to order table (PricePerUnit * Qty) | High | BRD | Data Processing
| Data Transformation | US-002 | repo/etl/transform_order.py | TS-002 | TC-002 | Not Started | BR-001 |
Calculation logic error | |

| BR-003 | Remove Null, duplicate records from both tables | High | BRD | Data Quality | Data Cleaning
| US-003 | repo/etl/clean_data.py | TS-003 | TC-003 | Not Started | BR-001 | Incomplete clean-up | |

| BR-004 | Create ordersummary table if not exists | High | BRD | Data Modeling | Table Creation |
US-004 | repo/db/create_table.py | TS-004 | TC-004 | Not Started | BR-001 | Table creation failure | |

| BR-005 | Join customer and order data, load to SCD type 2 ordersummary | High | BRD | Data
Processing | SCD2 Logic | US-005 | repo/etl/switch_scd2.py | TS-005 | TC-005 | Not Started | BR-004 |
SCD logic defects | |

| BR-006 | Update ordersummary SCD2 on customer table change, maintain history | High | BRD | Data
Processing | SCD2 Update | US-006 | repo/etl/update_scd2.py | TS-006 | TC-006 | Not Started |
BR-005 | Missing history updates | |

| BR-007 | Create customeraggregatespend table (if not exists) | High | BRD | Data Modeling | Table
Creation | US-007 | repo/db/create_table.py | TS-007 | TC-007 | Not Started | BR-005 | Table creation
failure | |

| BR-008 | Aggregate TotalAmount from ordersummary; group by Name and Date; load to
customeraggregatespend | High | BRD | Data Reporting | Aggregation Logic | US-008 |
repo/etl/aggregate_spend.py | TS-008 | TC-008 | Not Started | BR-007 | Missing aggregates | |

| BR-009 | Implement all steps using delta live tables | High | BRD | ETL Architecture | Delta Live Tables
Orchestration | US-009 | repo/etl/live_tables.py | TS-009 | TC-009 | Not Started | All above steps | DLT
orchestration failure | |
